

Figure 3.1: Equivalent cylindrical thermal circuit used to compute the analytical reference heat flow for Mode C (Mixed mode, $h_\infty > 0$, $\varepsilon > 0$).

mistakes in the code. The three thermal modes were:

1. Mode A: External convection only, $(h_\infty > 0,\ \varepsilon = 0)$
2. Mode B: Thermal radiation only, $(h_\infty = 0,\ \varepsilon > 0)$
3. Mode C: Combined convection and radiation $(h_\infty > 0,\ \varepsilon > 0)$

For each mode, the analytical reference heat flow is computed using a thermal circuit approach (Figure 3.1) for which the verification case was designed to best-match the assumptions a thermal circuit would use. Those assumptions are:

- **Axisymmetric Geometry:** both sections of pipe are straight, co-linear, and cylindrical.
- **Constant Centerline Temperature:** the non-adiabatic region being examined is short and the flow is fast, resulting in a negligible ΔT from the region's inlet to outlet.
- **Steady-State Flow:** with constant boundary conditions, there is no surviving d/dt after the flow develops
- **Constant Material Properties:** the simulation and circuit use constant properties.
- **Constant Environmental Conditions:** the ambient temperature T_∞, convective heat transfer coefficient h_∞ and environmental emissivity (assumed equal to 1), are set as constants.